25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Eigenvalues, Eigenvectors & Diagonalization Practice Exercises

1 MCQ

Find $\|\mathbf{u}\|$ for $\mathbf{u} = (-9, -12)$.

A 21
B 14
C 15
D 225

2 MCQ

A 3×3 symmetric matrix Q_k has eigenvalues $\lambda_1 = 12$, $\lambda_2 = 3$, and $\lambda_3 = 14$. Determine the trace $\text{tr}(Q_k)$.

A 26
B 32
C 29
D 41

3 MCQ

A linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, find $T(\mathbf{v})$.

A $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$
B $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$
C $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$
D $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

4 MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If a point $P = (x, y)$ is transformed by A to $P' = (x', y')$, such that $\begin{pmatrix} x' \\ y' \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix}$, what is the image of the point $(3, 1)$ under this transformation?

A (7, 6)
B (6, 7)
C (8, 5)
D (5, 8)

5 MCQ

Consider two vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

A 1
B 0
C -7
D 25

6 MCQ

Consider the following system of linear equations represented by the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

A 1

B 2

C 3

D 4

7 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$. Calculate the magnitude of $\mathbf{v}$, denoted as $\|\mathbf{v}\|$.

A 13

B 17

C 7

D 169

8 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$. If this vector is plotted on a standard xy-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?

A The vector's direction in the xy-plane.

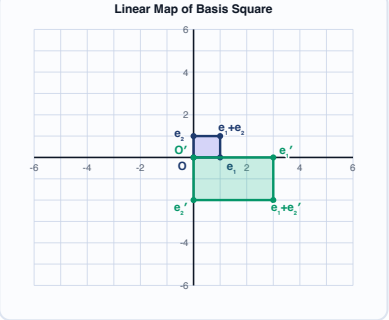
B The angle the vector makes with the positive x-axis.

C The distance of the point (3,4) from the origin.

D The sum of the components of the vector.

9 MCQ

Consider a linear transformation represented by the diagonal matrix $A = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is an eigenvector of A, what is the corresponding eigenvalue?



A 3

B -2

C 1

D 5

10 MCQ

Consider the upper-triangular matrix $T = \begin{pmatrix} 7 & 21 \\ 0 & 4 \end{pmatrix}$. Determine the eigenvalue product $\det(T) = \lambda_1 \lambda_2$.

A 31

B 7

C 28

D 35

11 MCQ

Consider a 2×2 matrix A with an eigenvalue $\lambda = 3$ and a corresponding eigenvector $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. If the trace of A is 7, find the other eigenvalue of A.

A 1

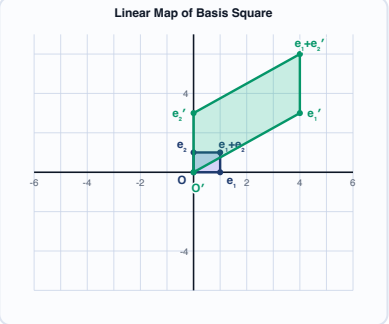
B 4

C 10

D 21

12 MCQ

Find the eigenvalues of $\begin{bmatrix} 4 & 0 \\ 3 & 3 \end{bmatrix}$.



A $3, 4$

B $0, 4$

C $4, 5$

D $-3, -4$

13 MCQ

A 2×2 matrix has eigenvalues $-1, -5$. Find its trace.

A 6

B 5

C -6

D -5

14 MCQ

Find the eigenvalues of $\begin{bmatrix} 4 & 0 & 0 \\ -2 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}$.

A $0, 4$

B $1, 2, 5$

C $0, 1, 4$

D $0, -1, -4$

15 MCQ

A 2×2 matrix has eigenvalues $5, 0$. Find its trace.

A 5

B 0

C 6

D -5

16 MCQ

Find the eigenvalues of $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 5 & 3 \\ 0 & 0 & 0 \end{bmatrix}$.

A $-2, -1$

B $0, 1, 6$

C $1, 0, -5$

D $-1, 0, 5$

17 MCQ

A 2×2 matrix has eigenvalues $6, -5$. Find its trace.

A -30

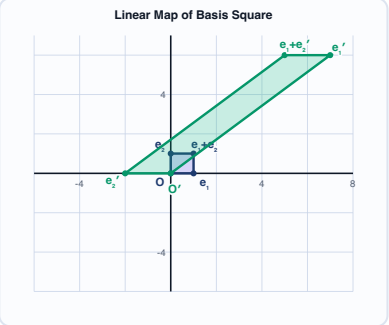
B 2

C 1

D -1

18 MCQ

$\mathbf{v} = \langle 4, 6 \rangle$ is an eigenvector of $A = \begin{bmatrix} 7 & -2 \\ 6 & 0 \end{bmatrix}$. What is its eigenvalue?



A 3

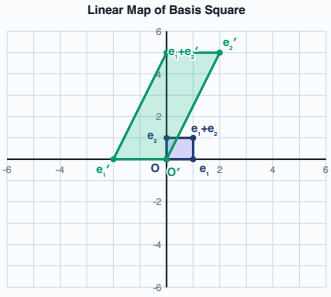
B 5

C 4

D -4

19 MCQ

Find the eigenvalues of $\begin{bmatrix} -2 & 2 \\ 0 & 5 \end{bmatrix}$.



A $2, -5$

B $-2, 2$

C $-1, 6$

D $-2, 5$

20 MCQ

A 3×3 matrix has eigenvalues $-5, 2, 2$. Find its determinant.

A -1

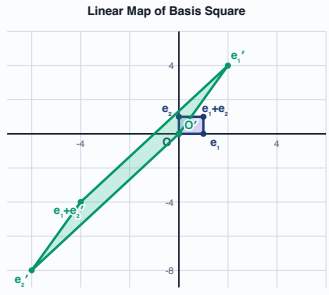
B -19

C -20

D 20

21 MCQ

$\mathbf{v} = \langle -6, -4 \rangle$ is an eigenvector of $A = \begin{bmatrix} 2 & -6 \\ 4 & -8 \end{bmatrix}$. What is its eigenvalue?



A -4

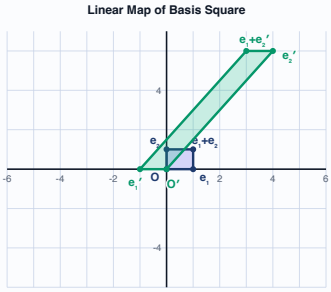
B -1

C 2

D -2

22 MCQ

Calculate the eigenvalues of $\begin{bmatrix} -1 & 4 \\ 0 & 6 \end{bmatrix}$.



A $1, -6$

B $-1, 4$

C $-1, 6$

D $0, 7$

23 MCQ

A 2×2 matrix has eigenvalues $6, -4$. Find its trace.

A 2

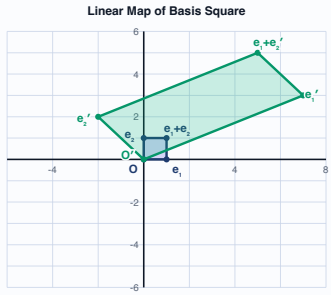
B 3

C -2

D -24

24 MCQ

$\mathbf{v} = \langle -4, -6 \rangle$ is an eigenvector of $A = \begin{bmatrix} 7 & -2 \\ 3 & 2 \end{bmatrix}$. What is its eigenvalue?



A 5

B 4

C -4

D -40

25 MCQ

Find the eigenvalues of $\begin{bmatrix} 6 & 0 & 0 \\ -1 & -4 & 0 \\ -3 & 2 & 6 \end{bmatrix}$.

A $-4, 6$

B $0, 6$

C $-3, 7$

D $4, -6$